

一维定常绝热等熵流

$$\frac{T}{T_0} = (1 + \frac{\gamma-1}{2} M^2)^{-1}$$

$$\frac{T}{T_0} = \frac{\gamma+1}{2 + (\gamma-1)M^2}$$

$$\frac{p}{p_0} = (1 + \frac{\gamma-1}{2} M^2)^{-\frac{\gamma}{\gamma-1}}$$
$$\frac{\rho}{\rho_0} = (1 + \frac{\gamma-1}{2} M^2)^{-\frac{1}{\gamma-1}}$$

$$\frac{p}{p^*} = (\frac{\gamma+1}{2 + (\gamma-1)M^2})^{\frac{\gamma}{\gamma-1}}$$
$$\frac{\rho}{\rho^*} = (\frac{\gamma+1}{2 + (\gamma-1)M^2})^{\frac{1}{\gamma-1}}$$

最大速度 (理论)

$$V_{max} = \sqrt{2h_0} = \sqrt{\frac{2}{\gamma-1} a_0^2}$$
$$\frac{V_{max}}{a_0} = \sqrt{\frac{2}{\gamma-1}}$$

如果 p_0, T_0 是确定的, 一旦最小截面出现临界状态, Q_m 就是最大值不变了 (堵塞流量, 堵塞流量)

临界状态与滞止

$$\frac{T_0}{T^*} = \frac{\gamma+1}{2}$$
$$\frac{p_0}{p^*} = (\frac{\gamma+1}{2})^{\frac{\gamma}{\gamma-1}}$$
$$\frac{\rho_0}{\rho^*} = (\frac{\gamma+1}{2})^{\frac{1}{\gamma-1}}$$

亚音速 (求 Q_m, M_{0e})

空气高压大罐 $p_0=1.0\text{MPa}$, $T_0=350\text{K}$

全管等熵流, 参考状态参数相同

$T_1 = T_0 = 320\text{K}$, $p_1 = \frac{p_0}{R T_1} = 7.99 \text{ kg/m}^3$, $V_1 = M_1 \sqrt{\gamma R T_1} = 244 \text{ m/s}$

$\frac{A_1}{A^*} = \frac{1}{M_1} (\frac{2 + (\gamma-1)M_1^2}{\gamma+1})^{\frac{\gamma+1}{2(\gamma-1)}} = 1.110$, $A^* = 4.32 \times 10^{-4} \text{ m}^2$, $A_1 = 4.80 \times 10^{-4} \text{ m}^2$

$\frac{A_2}{A^*} = \frac{1}{M_2} (\frac{2 + (\gamma-1)M_2^2}{\gamma+1})^{\frac{\gamma+1}{2(\gamma-1)}} = 2.317$, $M_2 = 0.26$

$Q_m = 0.04042 \frac{p_0}{\sqrt{T_0}} A^* = 0.933 \text{ kg/s}$

求熵增, 质量流量

Air $p_1=300\text{kPa}$, $T_1=60^\circ\text{C}$, $V_1=183\text{m/s}$

求: $p_2, T_2, \rho_2, S_2-S_1$

$a_1 = \sqrt{\gamma R T_1} = 366 \text{ m/s}$, $M_1 = V_1 / a_1 = 0.5$

$\frac{p_2}{p_1} = (1 + \frac{\gamma-1}{2} M_1^2)^{-\frac{\gamma}{\gamma-1}} = 0.843$, $p_2 = 415 \text{ kPa}$

$\frac{T_2}{T_1} = (1 + \frac{\gamma-1}{2} M_1^2)^{-1} = 0.952$, $T_2 = 350 \text{ K}$

$\frac{\rho_2}{\rho_1} = (1 + \frac{\gamma-1}{2} M_1^2)^{-\frac{1}{\gamma-1}} = 0.361$, $\rho_2 = 139 \text{ kg/m}^3$

$\frac{T_2}{T_2^*} = (1 + \frac{\gamma-1}{2} M_2^2)^{-1} = 0.747$, $T_2^* = 350 \text{ K}$, $T_2 = 262 \text{ K}$

$S_2 - S_1 = c_p (\ln \frac{p_2}{p_1} - \ln \frac{\rho_2}{\rho_1}) = \frac{\gamma R}{\gamma-1} \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} = 24.1 \text{ J/(kg}\cdot\text{K)}$

正与斜激波

$$M_2 = \frac{1 + [(\gamma-1)/2] M_1^2}{\gamma M_1^2 - (\gamma-1)/2}$$

$$\frac{\rho_2}{\rho_1} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2} = \frac{(\gamma+1) M_1^2}{2 + (\gamma-1) M_1^2} > 1$$

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) > 1$$

$$\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right] \frac{2 + (\gamma-1) M_1^2}{(\gamma+1) M_1^2} > 1$$

$$\frac{A_1}{A_2} = \frac{p_{02}}{p_{01}} = \frac{p_2}{p_1} \frac{\rho_2}{\rho_1} \frac{T_2}{T_1} = \left(\frac{2 + (\gamma-1) M_1^2}{\gamma M_1^2 - (\gamma-1)/2} \right)^{\frac{\gamma}{\gamma-1}} \left(\frac{2\gamma}{\gamma+1} M_1^2 - \frac{\gamma-1}{\gamma+1} \right)^{-\frac{1}{\gamma-1}} < 1$$

$$\frac{p_{02}}{p_{01}} = \frac{p_2}{p_1} \frac{\rho_2}{\rho_1} \frac{T_2}{T_1} = \left(\frac{\gamma+1}{4\gamma M_1^2 - 2(\gamma-1)} \right)^{\frac{\gamma}{\gamma-1}} \frac{1 - \gamma + 2\gamma M_1^2}{\gamma+1}$$

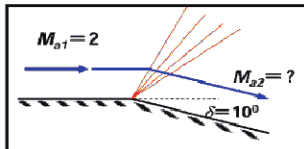
$$\tan \delta = \frac{2 \cot \beta (M_1^2 \sin^2 \beta - 1)}{(\gamma+1) M_1^2 - 2(M_1^2 \sin^2 \beta - 1)}$$

$$\text{将正激波关系中 } M_1 \text{ 用 } M_1 \sin \beta \text{ 代替}$$

$$M_2 \text{ 用 } M_2 \sin(\beta - \delta) \text{ 代替}$$

正激波 (相对激波)

$$M_1 > 1, M_2 < 1$$



Prandtl-Meyer角 θ

$$\theta = \sqrt{\frac{\gamma+1}{\gamma-1}} \arctan \sqrt{\frac{\gamma-1}{\gamma+1} (M^2 - 1)} - \arctan \sqrt{M^2 - 1}$$

θ 为气流从 $M=1$, $\theta=0$ 到 M 流动的偏转角

$M_1=2$

查表 $\theta = 26.5^\circ$ (相对 $M_0=1$ 的偏转)

加速到 M_2 , 所以 $\theta_2 > \theta_1$, 扰动前后气流方向 $\Delta\theta = \delta = 10^\circ$

$$\theta_2 = \theta_1 + \delta = 26.5^\circ + 10^\circ = 36.5^\circ$$

则查表 $M_2 = 2.39$

整个偏转过程是加速 ($M_2 > M_1$), 等熵膨胀过程, 流动参数可用等熵关系

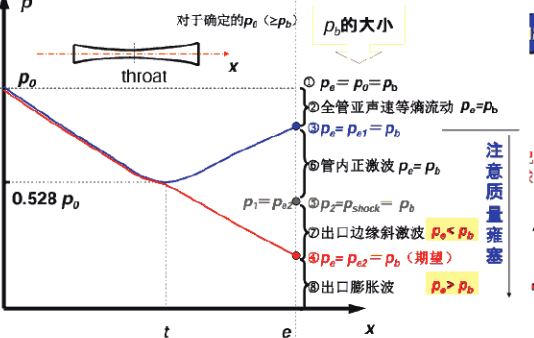
例题: $M_1=3$, 求 $M_2=?$, $M_3=?$

基本分析判断: $\theta_2 = \theta_3 = \theta = 10^\circ$

图或公式: $M_1=3$ 斜激波角 $\beta_1=28^\circ$, 利用正激波关系 $M_{2n} = M_2 \sin(\beta_1 - \theta_2) = 0.7355$, $M_2 = 2.38$

图或公式: $M_2=2.38$ 斜激波角 $\beta_2=33^\circ$, 利用正激波关系 $M_{3n} = M_3 \sin(\beta_2 - \theta_3) = 0.7860$, $M_3 = 2.011$

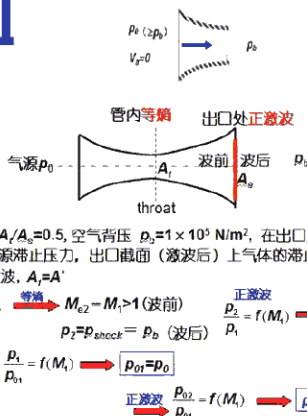
几何条件 A_2/A_1 决定了三种可能的特征流动压力



Laval 喷管

- $p_0 > p^*$ 出口处不会达到声速流动, 全管亚声速, 且 $p_e = p_b$
- $p_0 < p^*$ 出口处 (最小截面) 达到声速流动, $p_e = p^*$, 管内亚声速

质量流量



例题

$$M_1 = 1.8, \theta_1 = 20.73^\circ$$

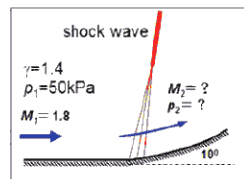
$$\text{气流 } \theta_2 = 20.73^\circ - 10^\circ = 10.73^\circ$$

$$\text{则 } M_2 = 1.46$$

$$\frac{p_2}{p_1} = \frac{p_2}{p_0} \frac{p_0}{p_1} = f_1(M_2) f_1(M_1) = 0.28856 \times \frac{1}{0.17404} = 1.658$$

$$p_2 = 82.9 \text{ kPa}$$

减速等熵压缩升压过程



Laval管内激波位置的确定

已知 $A_2/A_1, p_{01}, p_{02}$

判断过程: 求出出口特征压力 p_{02}

$A_2/A_1 \rightarrow$ 双解 $M_1 < 1$ 或 $M_1 > 1$

如果 $p_{shock} < p_b < p_{01}$ 则管内有激波

要确定激波位置, 求出 $A_2/A_1 = A_2^*/A_1^* = f(M)$, 关键是波前 M_1

$\frac{A_2}{A_1} = \frac{A_2^*}{A_1^*} = \frac{A_2^*}{A_1^*} \frac{p_{01}}{p_{02}} = \frac{A_2^*}{A_1^*} \frac{p_{01}}{p_{02}} \frac{p_{02}}{p_{01}} = \frac{A_2^*}{A_1^*} \frac{p_{01}}{p_{02}}$

$\frac{A_2}{A_1} = \frac{1}{M_2} (\frac{2 + (\gamma-1)M_2^2}{\gamma+1})^{\frac{\gamma+1}{2(\gamma-1)}}$

$\frac{p_{02}}{p_{01}} = (1 + \frac{\gamma-1}{2} M_1^2)^{-\frac{\gamma}{\gamma-1}}$

利用波后等熵和出口 M_2 有关

分析定常流

p_0, T_0, ρ_0, V_0 滞止状态

收缩管内是加速, 降压过程

分析环境背压 $p_b \sim p^*$ 的相对大小

- 若 $p_b > p^*$ 若环境压力高于临界压力时, 出口处不会达到临界声速流动, 全管连续等熵亚声速流动, 且 $p_e = p_b$
- 若 $p_b = p^*$ 出口处 (最小截面) 达到声速流动, 管内连续等熵亚声速
- 若 $p_b < p^*$ 管内流动及出口状态同上完全一样, $p_e = p^*, M_{0e} = 1$

因为 $p_b > p^*$, 管内气流降压膨胀不足 (欠膨胀), 出口后继续加速、膨胀降压

质量流量达到最大值

例题

已知: $A_2/A_1=0.5$, 空气背压 $p_b=1 \times 10^5 \text{ N/m}^2$, 在出口处恰好正激波

求: 气流滞止压力, 出口截面 (波后) 上气体的滞止压力。

有激波, $A_2/A_1 = A_2^*/A_1^* = f(M_1)$

$A_2/A_1 \rightarrow$ 等熵 $M_2 = M_1 > 1$ (波前) $\frac{p_2}{p_1} = f(M_1) \rightarrow p_1$

$p_2 = p_{shock} = p_b$ (波后) $\frac{p_2}{p_1} = f(M_1) \rightarrow p_1$

等熵 $\frac{p_1}{p_{01}} = f(M_1) \rightarrow p_{01} = p_0$

正激波 $\frac{p_{02}}{p_{01}} = f(M_1) \rightarrow p_{02}$

Laval 喷管流动, 已知 $A_2/A_1=2.0, p_0=400 \text{ kN/m}^2$

问背压 $p_b=100 \text{ kN/m}^2$, $p_1=22 \text{ kN/m}^2$ 对应的流动状态, 并求激波角、偏转角?

解: $M_2=2.2, p_{02}=0.0935 \times 400 = 37.4 \text{ kN/m}^2$

Or: $p_1 = p_{02}$

$\frac{p_2}{p_1} = 5.48$ (正激波关系式)

$\frac{p_2}{p_1} = 5.48$

$p_2 = p_{shock} = 205 \text{ kN/m}^2$

$p_b < p_{02}$ 膨胀波 (P-M 流)

$\frac{p_b}{p_0} = \frac{22}{400} = 0.055$ (等熵查表)

$M_2 = 2.54 \rightarrow v(M_2) = 40.05^\circ$

又 $M_1 = 2.2 \rightarrow v(M_1) = 31.73^\circ$

$\delta = v(M_2) - v(M_1) = 8.32^\circ$

3. (20分) 如图飞行器 LBC 在超声速风洞 (拉伐尔喷嘴) 中做吹风实验, 投影实验可以看见在飞行器的 LB 下端有一道较强的斜激波 BA, 斜激波与来流方向夹角为 55° , 假设来流空气 (为理想气体) 比热完全气体, 不计重力, $LB \perp LC = 1m$, $\angle LBC = \angle CBO = 1/2 \angle LBO$, 已知风洞 Ae/At=2.005, $P_0=600kPa$, 试求该飞行器垂直纸面单位宽度的阻力和升力。 (提示: 风洞实验段入口 Ae 截面刚好超声速流动)

$$\frac{A_e}{A_t} = 2.005, \quad M_{a_{e1}} = 0.3, \quad \frac{P_{e1}}{P_0} = 0.93947, \quad P_{e1} = 5.6365 \times 10^5 Pa$$

$$M_{a_{e2}} = 2.2, \quad \frac{P_{e2}}{P_0} = 0.09352, \quad P_{e2} = 5.6112 \times 10^4 Pa$$

$$P_1 = P_{e2}, \quad \beta = 55^\circ, \quad Ma = 2.2 \Rightarrow \delta = 23.84^\circ$$

$$M_{1n} = M_1 \sin \beta = 1.8021, \quad M_{2n} = 0.6165, \quad M_2 = \frac{M_{2n}}{\sin(\beta - \delta)} = 1.1915$$

$$\frac{P_2}{P_1} = 3.6133 \Rightarrow P_2 = 2.0275 \times 10^5 Pa, \quad \frac{P_2}{P_{02}} = 0.41778, \quad P_{02} = 4.853 \times 10^5 Pa$$

$$M_1 = 2.2 \Rightarrow \nu(M_1) = 32.0^\circ, \quad \nu_{new} = \nu(M_1) + \left(\frac{23.84}{2}\right)^\circ = 43.92^\circ, \quad M_3 = 2.718$$

$$\frac{P_3}{P_1} = \left[\frac{1 + \frac{\gamma-1}{2} M_3^2}{1 - \frac{\gamma-1}{2} M_1^2} \right]^{\frac{\gamma}{\gamma-1}} = 0.4467, \quad P_3 = 2.5065 \times 10^4 Pa$$

$$M_2 = 1.1915 \Rightarrow \nu(M_2) = 3.5^\circ, \quad \nu_{new2} = \nu(M_2) + 23.84^\circ = 27.35^\circ, \quad M_4 = 2.041$$

$$\frac{P_4}{P_2} = \left[\frac{1 + \frac{\gamma-1}{2} M_4^2}{1 - \frac{\gamma-1}{2} M_2^2} \right]^{\frac{\gamma}{\gamma-1}} = 0.2876, \quad P_4 = 5.8301 \times 10^4 Pa$$

$$F_{lift} = P_2 A_2 \cos(\delta) + P_4 A_4 - P_3 A_3 \cos(\delta) = 2.2083 \times 10^5 N$$

$$F_{drag} = P_2 A_2 \sin(\delta) - P_3 A_3 \sin\left(\frac{\delta}{2}\right) = 7.6771 \times 10^4 N$$

2. (15分) 空气通过渐缩喷嘴排向大气, 大气压 $p_a = 100kPa$, 进口 $A_1 = 0.558m^2$, $\rho_1 = 150kg/m^3$, $T_1 = 300K$, 出口为亚音速, $A_2 = 0.2m^2$, $R = 287 J/kg \cdot K$, 假设流动为无粘常比热完全气体的绝热流动, 忽略重力。求: (1) 通过喷嘴的空气质量流量 $\dot{m}$; (2) 欲使喷嘴不动需施加多大的外力?

$$P_2 = P_1 = 100kPa, \quad \gamma = 1.4$$

$$C_p = \frac{\gamma}{\gamma-1} R = \frac{1.4}{1.4-1} \cdot 287 = 1004.5 J/kg \cdot K$$

由绝热可得

$$\left(\frac{T_2}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = 267.1834K$$

$$\rho_1 = \frac{P_1}{RT_1} = 1.7422 kg/m^3, \quad \rho_2 = 1.3041 kg/m^3$$

From Continuity Equation

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2 \Rightarrow \frac{V_2}{V_1} = \frac{\rho_1 A_1}{\rho_2 A_2} = 0.2683$$

From Energy Equation

$$e + \frac{p}{\rho} + \frac{V^2}{2} = const \Rightarrow h + \frac{V^2}{2} = const \Rightarrow \frac{V_2^2}{2} + c_p T_2 = \frac{V_1^2}{2} + c_p T_1$$

$$\Rightarrow V_2 = \sqrt{\frac{2c_p(T_1 - T_2)}{1 - \left(\frac{V_1}{V_2}\right)^2}} = 266.5381 m/s, \quad V_1 = 71.5122 m/s$$

From Momentum Equation

$$-\rho_1 A_1 V_1^2 + \rho_2 A_2 V_2^2 = F_x + P_1 A_1 - P_2 A_2$$

$$F_x = -50142.2376 N, \quad R_x = -F_x = 50.1422 kN$$

For it to not move

$$F'_x = -R_x + P_2(A_1 - A_2) = -14342.2376 N = -14.3422 kN$$

石油在长度 $L=50m$ 、直径 $d=100mm$ 的管道内流动, 管轴对于水平的倾斜度为 30° , $\mu=0.285 N \cdot s/m^2$, $\rho=950 kg/m^3$, $Re_{cr}=2000$ 试确定
(1) 为保持层流状态的最大允许流量;
(2) 相应的进出口压差。

解: (1) 若流动管内层流 (即Hagen-Poiseuille流)

$$Re_{cr} = \frac{\rho V d}{\mu} = 2000$$

$$\Rightarrow V = 6 m/s, \quad Q = V \frac{\pi d^2}{4} = 0.047 m^3/s$$

(2) 相应的进出口压差

$$\Delta p^* = \lambda \frac{1}{2} \rho V^2 \frac{L}{d} = \frac{64}{Re_{cr}} \frac{1}{2} \rho V^2 \frac{L}{d} = 273600 N/m^2$$

$$\Delta p^* = \Delta p + \rho g \Delta h$$

$$\Delta h = L \sin \alpha$$

$$\Delta p = 273600 - \rho g \Delta h = 40850 N/m^2$$

例题3: 无限长的半径为 R_1 的圆柱, 以常速度 U 沿轴线方向在无限长的半径为 R_2 的同轴圆柱面内运动, 试确定由此引起的两柱面间的粘性不可压缩流体运动的速度剖面 (质量力略去不计, 且已知轴向压力梯度为零)

解: 采用柱坐标 (r, z) 轴与轴线重合

由题意知: $V_r = V_\theta = 0, \quad V_z = V$

$$\frac{\partial}{\partial t} = 0, \quad \frac{\partial}{\partial \theta} = 0, \quad \frac{\partial p}{\partial z} = 0$$

由连续方程知: $\frac{\partial V}{\partial r} = 0 \Rightarrow V = V(r)$

由径向运动方程知: $\frac{\partial^2 p}{\partial r^2} = 0$ 周向运动方程自动满足

由轴向运动方程知: $\frac{d}{dr} \left(r \frac{dV}{dr} \right) = 0$

$$c_1 = -\frac{U}{\ln(R_2/R_1)}$$

$$c_2 = \frac{U \ln R_2}{\ln(R_2/R_1)}$$

积分两次得: $V = c_1 \ln r + c_2$

边界条件: $V|_{r=R_1} = U, \quad V|_{r=R_2} = 0$

所以: $V = U \frac{\ln(R_2/r)}{\ln(R_2/R_1)}$

一、圆管内不可压缩流体定常层流流动 (Hagen-Poiseuille流动) $Re < 2000-2300$

定常 $\frac{\partial}{\partial t} = 0$, 轴对称 $\frac{\partial}{\partial \theta} = 0, \quad V_\theta = 0$

充分发展流动, 速度 $\frac{\partial}{\partial z} = 0$

连续方程 $\nabla \cdot \vec{V} = 0$

柱坐标 (r, z)

$$\vec{V} = V_r \vec{e}_r + V_z \vec{e}_z$$

$$\frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_r}{\partial r} + \frac{\partial V_z}{\partial z} = 0$$

积分后

$$rV_r = c \quad (V_r \text{ 大小是有限的})$$

由 $r=0$, 确定 $c=0$

若 $c=0$, 一般 $r \neq 0, \therefore V_r = 0$

$\therefore$ 速度只有一个分量 $\vec{V} = V_z \vec{e}_z$ 而且 $V_z = f(r)$

不可压 N-S 方程

$$\frac{D\vec{V}}{Dt} = -\frac{\nabla p^*}{\rho} + \nu \nabla^2 \vec{V}$$

$$\frac{\partial}{\partial t} = 0, \quad \frac{\partial}{\partial \theta} = 0, \quad V_\theta = 0, \quad V_r = 0$$

速度充分发展 $\frac{\partial}{\partial z} = 0$

$$V_z = f(r) = ?$$

$$\frac{\partial V_z}{\partial t} + V_r \frac{\partial V_z}{\partial r} + V_z \frac{\partial V_z}{\partial z} = -\frac{1}{\rho} \frac{\partial p^*}{\partial z} + \nu \left(\frac{\partial^2 V_z}{\partial r^2} + \frac{1}{r} \frac{\partial V_z}{\partial r} \right)$$

非线性项自动消失 (为零)

$$\frac{\partial p^*}{\partial r} = 0 \quad (p^* \text{ 和 } r \text{ 无关})$$

$$\frac{\partial p^*}{\partial z} = \frac{\mu}{r} \frac{d}{dr} \left(r \frac{dV_z}{dr} \right)$$

对 r 积分

$$V_z = \frac{1}{4\mu} \frac{dp^*}{dz} r^2 + c_1 \ln r + c_2$$

通解

若速度边界条件: $(V_z)_{r=R} = 0, \quad (V_z)_{r=0} = \text{有限}$

管壁处速度无滑移 管中心速度有限

$$V_z = \frac{1}{4\mu} \frac{dp^*}{dz} (r^2 - \frac{d^2}{4})$$

单位管长度上的广义压差, 与 r 无关

$$p^* = p^*(z)$$

圆管内充分发展定常层流 H-P 流动

流量

$$Q = \int_0^d V_z 2\pi r dr = -\frac{\pi d^4}{128\mu} \frac{dp^*}{dz}$$

Hagen-Poiseuille 定律

平均速度

$$\bar{V} = \frac{Q}{\pi d^2/4} = -\frac{d^2}{32\mu} \frac{dp^*}{dz} = \frac{1}{2} V_{max}$$

(管内看成一维流动时, 各管截面用平均速度)

最大速度

$$(V_{max})_{r=0} = -\frac{d^2}{16\mu} \frac{dp^*}{dz}$$

广义压力, h 轴与重力反向

$$p^* = p + \rho gh$$

$$\Delta p^* = \Delta p + \rho g \Delta h$$

例2: 毛细管黏度计 (测 μ)

AB 间近似看成充分发展层流

$$Q = -\frac{\pi d^4}{128\mu} \frac{dp^*}{dx}$$

$$\frac{dp^*}{dx} = \frac{p_a + \rho g h_a - (p_b + \rho g h_b)}{L}$$

(做了近似)

$$p_a - p_b + \rho g (h_a - h_b) = \rho g H$$

$$\mu = \frac{\pi d^4 \rho g (H+L)}{128 Q L}$$

近似: 无限长, 至少 $L/d > 100$, 入口效应可不计了

定常, H 应恒定, 实际选取一段时间内平均高度近似

层流, d 是毫米量级, 流速很慢, $Re < 2000$

圆管, 细造毛细管, 横截面一定

二、无限大平板不可压缩流体定常层流流动 (Couette流动)

定常 $\frac{\partial}{\partial t} = 0$, 速度场 $\frac{\partial}{\partial x} = 0$, 足够宽 $\frac{\partial}{\partial y} = 0, \quad w = 0$, 不计 v

$$\nabla \cdot \vec{V} = 0$$

$$\frac{D\vec{V}}{Dt} = -\frac{\nabla p^*}{\rho} + \nu \nabla^2 \vec{V}$$

$$\frac{\partial V_x}{\partial t} + V_x \frac{\partial V_x}{\partial x} + V_y \frac{\partial V_x}{\partial y} + V_z \frac{\partial V_x}{\partial z} = -\frac{1}{\rho} \frac{\partial p^*}{\partial x} + \nu \left(\frac{\partial^2 V_x}{\partial y^2} + \frac{\partial^2 V_x}{\partial z^2} \right)$$

$$\frac{\partial V_x}{\partial t} + V_x \frac{\partial V_x}{\partial x} + V_y \frac{\partial V_x}{\partial y} + V_z \frac{\partial V_x}{\partial z} = -\frac{1}{\rho} \frac{\partial p^*}{\partial x} + \nu \frac{\partial^2 V_x}{\partial y^2}$$

上述流动特点的求解

$$u = \frac{1}{2\mu} \frac{dp^*}{dx} y^2 + c_1 y + c_2$$

若边界条件是 $(u)_{y=0} = 0, \quad (u)_{y=H} = U$

$$u = \frac{1}{2\mu} \frac{dp^*}{dx} (y^2 - Hy) + \frac{Uy}{H}$$

无限大平板充分发展层流多种类型

按上“动”下“静”固壁速度边界条件

$$u = \frac{1}{2\mu} \frac{dp^*}{dx} (y^2 - Hy) + \frac{Uy}{H}$$

取流动方向为 X 方向 $\frac{dp^*}{dx}$ 是常数 $\frac{dp^*}{dx} = \frac{P_{x+\Delta x} - P_x}{\Delta x}$ $p^* = p + \rho gh$

取 h 方向为重力的反方向

$$\tau = (\mu_{yx})_{wall} = \left(\mu \frac{du}{dy} \right)_{y=0} = -\frac{1}{2} \frac{dp^*}{dx} H + \mu \frac{U}{H}$$

壁面切应力

充满油的缸缸内有长 L 活塞向下以 V_0 等速运动, 油通过间隙 δ 向上流动, $\delta \ll d$, 流动是层流, 已知油粘度系数 μ , 密度 ρ , 活塞上下压差 $p_2 - p_1$, 活塞直径 d , 间隙 δ , 忽略端部效应, 求 V_0 。

解: $\delta \ll d, \quad b = \pi d, \quad H = \delta$ 环形通道, 缝隙内看成无限大两平板间的 Couette 流动

活塞运动排挤的油流量 = 缝隙中的流体流量

思路! $V_0 \frac{\pi d^2}{4} = \pi \int_0^\delta u dy$

$$u = \frac{1}{2\mu} \frac{dp^*}{dx} y^2 + c_1 y + c_2$$

从 Couette 流动通解入手

$$\begin{cases} (u)_{y=0} = 0 \\ (u)_{y=\delta} = -V_0 \end{cases} \rightarrow \text{定 } c_1 \text{ 和 } c_2$$

其中 $\frac{dp^*}{dx} = \frac{\Delta p + \rho g h}{L} = \frac{(p_2 + \rho g L) - p_1}{L} = \frac{p_2 - p_1}{L} + \rho g$

积分整理后得到

$$V_0 \frac{\delta^2}{2} + \frac{d^2}{4} = -\frac{\delta^3}{12\mu L} (p_2 - p_1 + \rho g L)$$

$$V_0 = \frac{\delta^3}{3\mu(2\delta + d)} \left(\frac{p_1 - p_2 - \rho g L}{L} \right) \approx \frac{\delta^3}{3\mu d L} (p_1 - p_2 - \rho g L)$$

灰尘 $\rho_1 = 2500 kg/m^3$, 空气 $\rho = 1.225 kg/m^3$, $\mu = 1.825 \times 10^{-5} m^2/s$, 求符合 Stokes 公式的灰尘的最大直径和沉降速度 V_0 。

$$\vec{F}_D = 6\mu a V_0 \vec{k} + \frac{4}{3} \rho g \pi a^3 \vec{e}_n$$

$$F_D = 6\mu a V_0 + \frac{4}{3} \rho g \pi a^3 = \frac{4}{3} \rho g \pi a^3$$

$$6\mu a V_0 = \frac{4}{3} (\rho_1 - \rho) g \pi a^3$$

$$\frac{3\mu^2 \pi 2a V_0}{\rho} = \frac{4}{3} (\rho_1 - \rho) g \pi a^3$$

$$\frac{3\mu^2 \pi}{\rho} Re = \frac{4}{3} (\rho_1 - \rho) g \pi a^3$$

取 $Re = 0.1$, 解得 $a = 13.6 \mu m, \quad d = 2a = 27.2 \mu m$

$$V_0 = \frac{Re \mu}{d \rho} = 0.035 m/s$$

3. 平面点涡 (Vortex自由涡)

在 (x_0, y_0) 处有自由涡, 逆时针为正

$$\Gamma = 2\pi V_\theta \sqrt{(x-x_0)^2 + (y-y_0)^2}$$

$$V_\theta = \frac{\Gamma}{2\pi \sqrt{(x-x_0)^2 + (y-y_0)^2}}$$

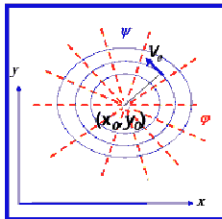
$$u = -V_\theta \sin \theta = -\frac{\Gamma}{2\pi} \frac{(y-y_0)}{(x-x_0)^2 + (y-y_0)^2}$$

$$v = V_\theta \cos \theta = \frac{\Gamma}{2\pi} \frac{(x-x_0)}{(x-x_0)^2 + (y-y_0)^2}$$

平面, 不可压, 无旋

$$\varphi = \frac{\Gamma}{2\pi} \arctg \frac{y-y_0}{x-x_0}$$

$$\psi = -\frac{\Gamma}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-y_0)^2}$$



$$(x_0=y_0=0) \quad V_\theta = \frac{\Gamma}{2\pi r}$$

$$\varphi = \frac{\Gamma}{2\pi} \theta$$

$$\psi = -\frac{\Gamma}{2\pi} \ln r$$

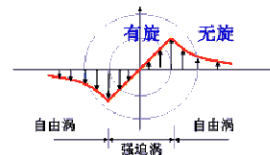
例题

实际涡更接近组合涡模型

组合涡: 核中心是一个强迫涡, 在核区之外的速度分布为自由涡。

$$V_\theta = \omega r \quad r \leq r_0$$

$$V_\theta = \frac{C}{r} \quad r > r_0$$



无旋可用Bernoulli方程, 写出压力分布 $(r \geq r_0)$

有旋, 用Euler方程, $\vec{v} \cdot \nabla \vec{v} = -\frac{\nabla p}{\rho}$, $\frac{V_\theta^2}{r} = \frac{1}{\rho} \frac{dp}{dr}$ ($r \leq r_0$)

在交接处, 速度连续, 压力连续

可确定C和压力分布

$$C = \omega r_0^2$$

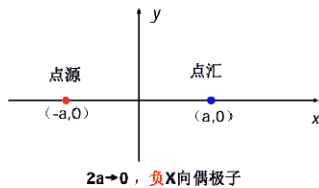
$$p = \frac{\rho \omega^2 r^2}{2} + p_c - \rho \omega^2 r_0^2, \quad (r \leq r_0)$$

$$p = p_c - \frac{1}{2} \frac{\rho \omega^2 r_0^4}{r^2} \quad (r \geq r_0)$$

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1. 偶极子 Doublet (source-sink Pair)

相距 $2a \rightarrow 0$ 的点源和点汇的组合, 有方向, 汇指向源



点源

$$(-a, 0) \quad \varphi = \frac{Q}{2\pi} \ln \sqrt{(x+a)^2 + y^2}$$

$$\psi = \frac{Q}{2\pi} \arctg \frac{y}{x+a}$$

点汇

$$(a, 0) \quad \varphi = -\frac{Q}{2\pi} \ln \sqrt{(x-a)^2 + y^2}$$

$$\psi = -\frac{Q}{2\pi} \arctg \frac{y}{x-a}$$

偶极子强度

$$\varphi = \lim_{a \rightarrow 0} \left[\frac{Q}{2\pi} \ln \sqrt{(x+a)^2 + y^2} - \frac{Q}{2\pi} \ln \sqrt{(x-a)^2 + y^2} \right]$$

$$= \frac{m}{2\pi} \frac{x}{x^2 + y^2}$$

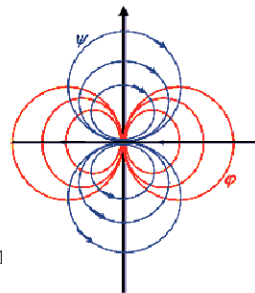
$$= \frac{m \cos \theta}{2\pi r}$$

$$\psi = \lim_{a \rightarrow 0} \left[\frac{Q}{2\pi} \arctg \frac{y}{x+a} - \frac{Q}{2\pi} \arctg \frac{y}{x-a} \right]$$

$$= -\frac{m}{2\pi} \frac{y}{x^2 + y^2}$$

$$= -\frac{m \sin \theta}{2\pi r}$$

流谱



$$\varphi = \frac{m}{2\pi} \frac{x}{x^2 + y^2} = \frac{m \cos \theta}{2\pi r}$$

$$\psi = -\frac{m}{2\pi} \frac{y}{x^2 + y^2} = -\frac{m \sin \theta}{2\pi r}$$

m —偶极子强度

2. 半无限长钝头体绕流

均匀来流

$$\varphi = U_\infty x$$

$$\psi = U_\infty y$$

点源

$$\varphi = \frac{Q}{2\pi} \ln r$$

$$\psi = \frac{Q}{2\pi} \theta$$

$$\varphi = U_\infty r \cos \theta + \frac{Q}{2\pi} \ln r = U_\infty r \cos \theta + U_\infty b \ln r$$

$$\psi = U_\infty r \sin \theta + \frac{Q}{2\pi} \theta = U_\infty r \sin \theta + U_\infty b \theta$$

$$V_r = \frac{\partial \varphi}{\partial r} = U_\infty \cos \theta + \frac{Q}{2\pi r}$$

$$V_\theta = \frac{1}{r} \frac{\partial \varphi}{\partial \theta} = -U_\infty \sin \theta$$

分析:

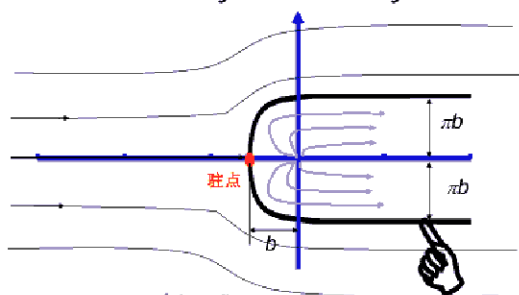
• 驻点位置 $V_r = U_\infty \cos \theta + \frac{Q}{2\pi r} = 0$
 $V_\theta = -U_\infty \sin \theta = 0$

驻点处流函数值 $\psi_0 = \pi b U_\infty$

过驻点的流线方程 $\psi = U_\infty r \sin \theta + \frac{Q}{2\pi} \theta = \pi b U_\infty$

由此得到过驻点的流线形状 $r = \frac{b(\pi - \theta)}{\sin \theta}$

$$\theta \rightarrow 0, y = \pi b; \quad \theta \rightarrow 2\pi, y = -\pi b;$$



过驻点流线 $r = \frac{b(\pi - \theta)}{\sin \theta}$, 或 $y = b(\pi - \theta)$ 可看做固体边界

对内流不感兴趣, 也不会影响外流, 所以组合解可以看成

半无限长钝头体无粘绕流

作业 17

3. 无环量无限长圆柱绕流

X向均匀来流

$$\vec{V} = U_\infty \vec{i}$$

$$\varphi = U_\infty x$$

$$\psi = U_\infty y$$

原点处-X向偶极子

$$\varphi = \frac{m \cos \theta}{2\pi r}$$

$$\psi = -\frac{m \sin \theta}{2\pi r}$$

$$\varphi = U_\infty x + \frac{m \cos \theta}{2\pi r} = U_\infty r \cos \theta + \frac{m \cos \theta}{2\pi r}$$

$$\psi = U_\infty y - \frac{m \sin \theta}{2\pi r} = U_\infty r \sin \theta - \frac{m \sin \theta}{2\pi r}$$

$$V_r = \frac{\partial \varphi}{\partial r} = U_\infty \left(1 - \frac{m}{2\pi U_\infty r^2}\right)$$

$$V_\theta = \frac{1}{r} \frac{\partial \varphi}{\partial \theta} = -U_\infty \left(1 + \frac{m}{2\pi U_\infty r^2}\right)$$

$$a^2 = \frac{m}{2\pi U_\infty}$$

$$\varphi = U_\infty \left(1 + \frac{a^2}{r^2}\right) r \cos \theta \quad V_r = U_\infty \left(1 - \frac{a^2}{r^2}\right) \cos \theta$$

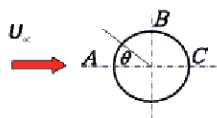
$$\psi = U_\infty \left(1 - \frac{a^2}{r^2}\right) r \sin \theta \quad V_\theta = -U_\infty \left(1 + \frac{a^2}{r^2}\right) \sin \theta$$

壁面上压力分布

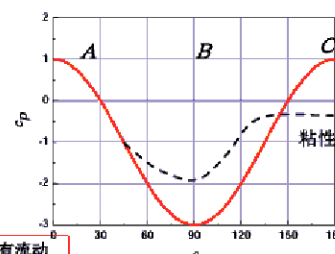
Bernoulli方程 $p_{r=a} = p_\infty + \frac{1}{2} \rho (U_\infty^2 - U_{r=a}^2)$

$$= p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$$

$$(C_p)_{r=a} = \frac{p_{r=a} - p_\infty}{\frac{1}{2} \rho U_\infty^2} = 1 - 4 \sin^2 \theta$$



• 没有考虑位势的变化
 有些问题必须考虑位势



粘性的真实流体压力分布不再对称而且有流动分离, 会产生阻力除摩擦之外的形阻。

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4. 有环量无限长圆柱绕流

均匀来流

$$\begin{aligned} \vec{V} &= U_\infty \vec{i} \\ \varphi &= U_\infty x \\ \psi &= U_\infty y \end{aligned}$$

-X向偶极子

$$\begin{aligned} \varphi &= \frac{m \cos \theta}{2\pi r} \\ \psi &= -\frac{m \sin \theta}{2\pi r} \end{aligned}$$

正向点涡

$$\begin{aligned} \varphi &= \frac{\Gamma}{2\pi} \theta \\ \psi &= -\frac{\Gamma}{2\pi} \ln r \end{aligned}$$

无环量无限长圆柱绕流

$$\begin{aligned} \varphi &= U_\infty \left(1 + \frac{a^2}{r^2}\right) r \cos \theta \\ \psi &= U_\infty \left(1 - \frac{a^2}{r^2}\right) r \sin \theta \end{aligned}$$

$$\begin{aligned} \varphi &= U_\infty \left(1 + \frac{a^2}{r^2}\right) r \cos \theta + \frac{\Gamma}{2\pi} \theta \\ \psi &= U_\infty \left(1 - \frac{a^2}{r^2}\right) r \sin \theta - \frac{\Gamma}{2\pi} \ln r \end{aligned}$$

$$V_r = U_\infty \left(1 - \frac{a^2}{r^2}\right) \cos \theta$$

$$V_\theta = -U_\infty \left(1 + \frac{a^2}{r^2}\right) \sin \theta + \frac{\Gamma}{2\pi r}$$

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圆球绕流— (注意来流与z轴平行, θ 是与z轴的夹角 $0 \sim \pi$)

$$\varphi = U_\infty \left(1 + \frac{a^3}{2R^3}\right) R \cos \theta$$

$$\psi = \frac{1}{2} \left(R^2 - \frac{a^3}{R}\right) \sin \theta$$

$$V_R = \frac{\partial \varphi}{\partial R} = U_\infty \left(1 - \frac{a^3}{R^3}\right) \cos \theta$$

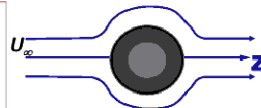
$$V_\theta = \frac{1}{R} \frac{\partial \varphi}{\partial \theta} = -U_\infty \left(1 + \frac{a^3}{2R^3}\right) \sin \theta$$

$$(V_R)_b = 0$$

$$(V_\theta)_b = -\frac{3}{2} U_\infty \sin \theta$$

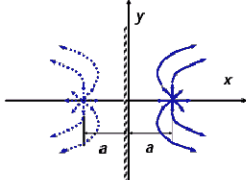
$$p_b = p_\infty + \frac{1}{2} \rho U_\infty^2 \left(1 - \frac{9}{4} \sin^2 \theta\right)$$

$$F_b = 0$$



**求压力没有考虑位势的变化

●点源+竖直墙壁



$$\psi_1 = \frac{Q}{2\pi} \arctg \frac{y}{x-a}$$

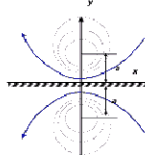
$$\psi_2 = \frac{Q}{2\pi} \arctg \frac{y}{x+a}$$

$$\psi = \psi_1 + \psi_2 = \frac{Q}{2\pi} \left(\arctg \frac{y}{x-a} + \arctg \frac{y}{x+a} \right)$$

验证 $x=0$ 是否是流线。 $x=0, \psi=0=\text{const}$, 是流线

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●点涡和墙壁



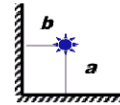
$$\psi_1 = \frac{\Gamma}{2\pi} \ln \sqrt{x^2 + (y-a)^2}$$

$$\psi_2 = -\frac{\Gamma}{2\pi} \ln \sqrt{x^2 + (y+a)^2}$$

$$\psi = \psi_1 + \psi_2 = \frac{\Gamma}{2\pi} \ln \sqrt{x^2 + (y-a)^2} - \frac{\Gamma}{2\pi} \ln \sqrt{x^2 + (y+a)^2}$$

验证 $y=0$ 是否是流线。 $y=0, \psi=0=\text{const}$ 是流线

墙角点源



解必须满足边界条件

$$\psi = \frac{Q}{2\pi} \left(\arctg \frac{y-a}{x-b} + \arctg \frac{y+a}{x-b} + \arctg \frac{y-a}{x+b} + \arctg \frac{y+a}{x+b} \right)$$

验证

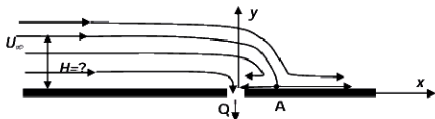
$x=0$ 是否是流线。 $x=0, \psi=0=\text{const}$ 是流线

$y=0$ 是否是流线。 $y=0, \psi=0=\text{const}$ 是流线

墙角是驻点

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2. 水以 1.5m/s 均匀流过一平面壁, 泵从一缝隙中以 $Q=0.01\text{m}^3/\text{s}$ (单位宽度) 的流量抽吸壁面的水, 求驻点位置; 求过驻点的流线方程; 问距离壁面 $H=?$ 多远的流体不会被吸入缝隙中?



解: 均匀流+点汇的例子 (解叠加, 等流函数为流线, 流函数之差为流量)

$$\begin{aligned} \varphi &= U_\infty x - \frac{Q}{\pi} \ln \sqrt{x^2 + y^2}, \quad \psi = yU_\infty - \frac{Q}{\pi} \arctg \frac{y}{x} \\ u = \frac{\partial \varphi}{\partial x} &= U_\infty - \frac{Q}{2\pi} \left(\frac{2x}{x^2 + y^2} \right), \quad v = \frac{\partial \varphi}{\partial y} = -\frac{Q}{2\pi} \left(\frac{2y}{x^2 + y^2} \right) \end{aligned}$$

$$\text{驻点 } (u=v=0, \text{ 驻点 A 位置}) \begin{cases} y=0 \\ x = \frac{Q}{\pi U_\infty} \end{cases} \quad (0, 2.12\text{mm})$$

求 H 两种方法:

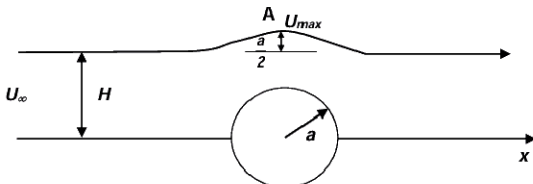
1. 过驻点 A 的流线的来流方向 $x \rightarrow -\infty$ 的高度为 H。 $U_\infty H = Q, \therefore H = \frac{Q}{U_\infty} = 6.66\text{mm}$

2. 计算驻点 A $(0, \frac{Q}{\pi U_\infty})$ 所在的流线上的流函数数值 (A 坐标代入流函数), 得到 $\psi=0$

$$y=0 \text{ 的负 x 轴的流线方程为 } \psi = yU_\infty - \frac{Q}{\pi} \arctg \frac{y}{x} = -Q$$

$$\text{流函数之差为流量, } 0 - (-Q) = U_\infty H, \therefore H = \frac{Q}{U_\infty} = 6.66\text{mm}$$

要做一个流线型物体, 其外形与无穷远来流绕 $r=a$ 圆柱流动的流场中一根流线形状相同, 如图所示, 问这根流线的相对圆柱对称轴的位置 $H=?$ 其上的最大速度与来流速度之比为多少?



解: 可直接用基本解或基本叠加解。

$$\psi = U_\infty \left(r - \frac{a^2}{r} \right) \sin \theta, \text{ 对于过 A 点 } \theta=\pi/2, r=H+a/2 \text{ 的流线方程, 可写为}$$

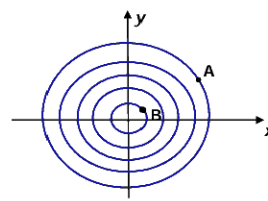
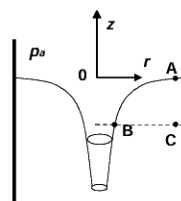
$$\psi = U_\infty \left(r - \frac{a^2}{r} \right) \sin \theta = U_\infty \left(H + \frac{a}{2} \right) \left[1 - \frac{a^2}{(H + \frac{a}{2})^2} \right]$$

X 轴和物面的流线流函数为零

$$\text{两流线之差为之间的流量 } U_\infty \left(H + \frac{a}{2} \right) \left[1 - \frac{a^2}{(H + \frac{a}{2})^2} \right] - 0 = U_\infty H, \therefore H = \frac{3}{2} a$$

$$\text{A 点 } \theta=\pi/2, r=H+a/2=2a, \quad V_r=0, V_\theta = -U_\infty \left(1 + \frac{a^2}{4a^2} \right) = -\frac{5U_\infty}{4}, U_{\max} = -V_\theta = \frac{5U_\infty}{4}, \quad \frac{U_{\max}}{U_\infty} = \frac{5}{4}$$

海底漩涡, 在远离中心的速度分布可以近似看成是一个平面点涡 $\varphi = \frac{\Gamma}{2\pi} \theta, \psi = -\frac{\Gamma}{2\pi} \ln r$, 要求确定自由面的形状。



$$v_\theta = \frac{1}{r} \frac{\partial \varphi}{\partial \theta} = \frac{\Gamma}{2\pi r}, \text{ 自由面上任一点 B 远离小孔, A 处 } r \rightarrow \infty, \text{ 速度为零静力学}$$

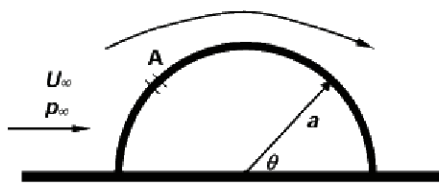
$$\frac{p_a}{\rho} + \frac{1}{2} v_a^2 = \frac{p_b}{\rho} + \frac{1}{2} v_b^2, \quad \frac{1}{2} v_b^2 + gz_b = 0, \quad z_b = z_s, \quad \frac{1}{2} \frac{\Gamma^2}{4\pi^2 r^2} + gz_s = 0$$

$$\therefore z_s = -\frac{\Gamma^2}{8g\pi^2 r^2}$$

漩涡不能终止在海水中 (因为终止处速度会无穷大), 所以会一直延伸到底底。(注意: 这里不是涡管, 是自由表面)

C 点压力 $p_c = p_s + \rho gh$, B 点 p_s , 所以漩涡会把海水中的东西吸过来

已知无穷远流速的风速为 U_∞ ，压力为 p_∞ ，绕过一个半径为 a ，长度为 L 的半圆柱型帐篷，设内部压力为 p_i ，用无粘理论推导由于内外压差帐篷受到的升力的表达式。如果风很大，在帐篷表面开个孔 A，使得内压和该处表面压力相等，问 A 孔对应的 $\theta=?$ 时使得帐篷受到的升力为零。（若是半球如何？）



$$p_{r-a} = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$$

$$F_y = p_i 2a + \iint -p_{r-a} dA \sin \theta = 2p_i a - \int_0^\pi \left[p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta) \right] \sin \theta a d\theta$$

$$F_y = 2ap_i - 2ap_\infty + \frac{5}{3} \rho a U_\infty^2$$

若开孔后升力为 0， $p_i = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$ ，代入 $F_y=0$ ，

$$\text{则 } \rho a U_\infty^2 (1 - 4 \sin^2 \theta) + \frac{5}{3} \rho a U_\infty^2 = 0. \therefore \sin^2 \theta = \frac{2}{3}, \theta = 54.7^\circ, 125.3^\circ$$

● 如果是半球体

$$p_{R-a} = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta)$$

$$F_y = p_i \pi a^2 + \iint -p_{r-a} \sin \theta \cos \varepsilon dA \quad dA = a^2 \sin \theta d\theta d\varepsilon \quad \theta=0 \sim \pi, \varepsilon=2(0 \sim \frac{\pi}{2})$$

$$\begin{aligned} F_y &= \pi a^2 p_i - 2 \int_0^\pi \int_0^{\frac{\pi}{2}} \left[p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta) \right] \sin \theta \cos \varepsilon a^2 \sin \theta d\theta d\varepsilon \\ &= \pi a^2 p_i - \pi a^2 \left(p_\infty + \frac{1}{2} \rho U_\infty^2 \right) + \rho U_\infty^2 \frac{9}{4} a^2 \int_0^\pi \cos \varepsilon d\varepsilon \int_0^\pi \sin^4 \theta d\theta \\ &= \pi a^2 p_i - \pi a^2 \left(p_\infty + \frac{1}{2} \rho U_\infty^2 \right) + \rho U_\infty^2 \frac{9}{4} a^2 \frac{3\pi}{8} \\ &= \pi a^2 p_i - \pi a^2 p_\infty + \frac{11}{32} \pi a^2 \rho U_\infty^2 \end{aligned}$$

若开孔后升力为 0， $p_i = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta)$ ，代入 $F_y=0$ ，

$$\text{则 } \frac{\pi a^2 \rho U_\infty^2}{2} (1 - \frac{9}{4} \sin^2 \theta) + \frac{11}{32} \pi a^2 \rho U_\infty^2 = 0. \therefore \sin^2 \theta = \frac{3}{4}, \theta = 60^\circ, 120^\circ$$

在 $\theta=60^\circ$ 、 120° 开孔升力为 0。

DNS – Direct Numerical Simulation 直接数值模拟

DES – De-attached Eddy Simulation 脱体涡数值模拟方法

LES – Large Eddy Simulation 大涡模拟

RANS – Reynolds-averaged Navier-Stokes equations

HWA – Hot-wire Anemometry

PIV – Particle Image Velocimetry

LDV – Laser Doppler Velocimetry

PDPA – Phase Doppler Particle Analysers

☐ A 理想流体也存在层流湍流之分

☐ B Re反映了粘性力与惯性力之比值

☒ 微管道层流到湍流可能会提前转换

☒ D 流体本构关系反映了流体的固有特性

☒ 粘性流体机械能一定是耗散的

☐ B 微管道从层流到湍流转换一般会延迟

☐ C Re数是粘性力和惯性力之比，粘性越大，Re越大。

☒ 粘性流体中涡能自生自灭

☒ A 无旋流动在非正压流场中可以变成有旋流动

☒ B 伯努利方程在定常、无粘，正压，质量力有势的无旋流场中任意点均成立

☒ C 对粘性流体流动可有流函数存在

☐ D 无粘、正压、质量力有势的无旋流体永远无旋

(✓) 重力场中一飞行器在不可压缩理想连续流场中由静止做加速运动，流场一定无旋。

(X) 不可压湍流的时均连续方程和动量方程构成封闭性方程组。

(X) 对于理想流体，其流动状态可以分为层流与湍流两种状态。

(✓) 粘性流体流动可有流函数存在。

(✓) 亚音速来流 Laval 喷管中管出口外欠膨胀流动，喷管的最小截面处一定是声速。

(✓) 正激波波后压力升高，密度升高，但总压、总温降低。

(✓) 静止正激波后气流速度一定是亚声速，而斜激波则不一定是。

(X) 流线型物体高 Re 绕流因为逆压梯度区小，不可能出现边界层分离。

(✓) 粘性流体和理想流体的微分形式的连续方程完全相同。

(✓) 边界层内粘性力大于惯性力，所以粘性不能忽略。

(X) 边界层的粘性外分界线就是流线。

(✓) 相同 Re 数下，相同管径和管长的粗糙管的沿程阻力可以与光滑管的相同。